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Final Report:
Running Shoes with Interchangeable Soles



**DARTMOUTH
ENGINEERING
THAYER SCHOOL**

Group 6:
Nathan Hammerschmitt Le Gal, Marina Rodriguez, Sam Li

MShop Instructors:
Mingwei Zhou, Cody Best, Lee Schuette

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Executive Summary

Running is a popular recreational activity favored by those seeking physiological and mental health benefits. However, its demanding nature can make it challenging to establish as a habit. The strain placed on a runner's legs is greatly influenced by their choice of footwear as shoes serve as the interface between the runner and the ground. Optimizing this interface can enhance a runner's performance and satisfaction while poor optimization can lead to subpar performance and increased risk of injury, hindering runners' ultimate goal of longevity and experiencing the associated benefits.

We identified two key areas where current running shoes fall short in promoting longevity: versatility and durability. Conventional running shoes typically offer a singular running feel and have a limited lifespan, necessitating frequent replacements. To overcome these limitations, some resort to owning multiple shoes with varying qualities and rotating them throughout the week. However, this approach is often financially inaccessible to many due to the high cost of running shoes. These issues are particularly pronounced among younger demographics, where the financial burden of footwear can pose a significant obstacle to establishing a running habit.

As a result, our target users primarily consist of younger age groups of American runners (20.1 million), although recreational runners of all ages could benefit from our solution. Our users are those who run frequently enough to experience significant shoe wear and be concerned about longevity, typically defined by mileage levels. Our target purchasers also overlap with these demographics (B2C), but we also see potential opportunities with casual running clubs, school teams, and independent running stores (B2B).

Our solution is a modular running shoe system consisting of two parts: an upper and a sole. Uppers have a longer lifespan as they are not subjected to the same impact forces as soles, which wear down due to repeated ground contact. The interchangeability of the sole allows runners to swap them according to their needs and durability concerns. We prioritized durability (lifetime mileage), versatility (use cases and performance metrics), and affordability (unit cost and annual cost) as the most important specifications based on the needs of younger runners aiming for consistent training and longevity. Our final prototype, a semi-functional 'works-like' model, successfully implemented a series of interlocking channels using a combination of existing shoe components and 3D-printed TPU locking mechanism parts.

While we have created a viable design, our next steps include altering the directionality of our mechanism, adding signifiers for clearer communication during attachment, developing a true running shoe upper, and incorporating additional securing features for a complete lock. We also plan to explore our secondary mechanism, a series of dovetail joints.

Our team drew upon data-based research, design thinking, entrepreneurship, and iterative prototyping skills to complete the project, and will continue to do so moving forward. We used our individual strengths and coordinated with mentors from the Machine Shop; we worked closely with Lee Schuette to learn the shoe-making process and coordinated with Cody Best, Mingwei Zhou, and Dr. Daniel Cullen to build, and test our prototype. The prototyping process taught us a great deal about the complex geometry of footwear, led us to acquire technical skills in assembling physical products, and made us competent in assessing our solution through analysis and user-testing. We plan to continue iterating on our current design; a long-term goal is to file a provisional patent application, and either create a new line of running shoes or sell the design to an established footwear company.

Problem and User

Magnitude and Scope

Running is a rewarding yet demanding activity; specifically, running can yield physiological and psychological benefits such as lowered resting heart rate and self-confidence, but the impact force placed upon runners' joints can range from 1.5 to 5 times their bodyweight [1]. For this reason, proper footwear can accentuate one's exercise experience by absorbing shock and supporting the legs [2]. Running shoes allow individuals to run farther and tire less, reaping greater amounts of proven benefits [3].

However, current running shoes fall behind in two major areas: versatility and durability. The current most optimal way to run is to possess many shoes, and a variety of shoes—this is called a “shoe rotation system.” When runners accrue miles on their footwear, the rubber outsole wears down, and the EVA or Pebax foam midsole becomes permanently compressed. This alteration in the physical structure of the shoe compromises its user experience [4]. Since the vast majority of running shoes last between 300 and 500 miles [5], using many shoes ensures runners are exposed to slight variations in biomechanics and never run on a worn shoe, thus improving the overall longevity of their running.

Runners must still, though, purchase new shoes, and median shoe replacement time is between 3 and 5 months [6]. Running shoe prices have also been on the rise [7], and while longevity in one's running is an issue that all runners face, younger recreational runners in high school, college, and in their early-career are confronted with the reality of buying 2 to 4 pairs of shoes a year. When 70+ Dartmouth Club Sports athletes were surveyed, 44% stated they only owned one pair of shoes, and many cited financial reasons as the determining factor:

“Running shoes are expensive, I only own one pair”

“Rotating shoes is too much work! And I don't put on enough miles to justify it. Also it's expensive.”

“Rotating shoes is expensive, and I find just one pair gets the job done.”

- Dartmouth Club Triathlon Team athletes

Furthermore, while elite and competitive runners have shoe rotation systems by default, shoe wear and replacement is an even greater issue, as a conventional, single pair might only last 6 weeks (with an assumed mileage per week of 50 to 80 miles) [8].

At a grand scale, 15% of the United States population participates in some form of running or jogging. These are 50 million Americans of which 50% will sustain some form of injury during the year [9, 10]. 10.9 million of these Americans also fall into the youth category (ages 6-17) [11] and 9.2 million fall into the young adult category (ages 18-24) [12]. These are runners that are just being introduced to the sport and at a stage of developing healthy habits—running longevity and price considerations are their top priority. 94.7% of Dartmouth Club Sports athletes cited “feel” as one of the most important factors in buying a shoe—it's the “good feel” runners seek from new shoes. Our solution would strive to address this need by providing consistent good feelings via increased durability and a wider selection of feelings through increased versatility. The economic outcome of the needs of such runners can be seen in the total athletic footwear market valuation, which, in 2022, was \$15.44 billion dollars [13].

Problem Statement

A singular pair of running shoes serves a singular purpose (limited versatility) and running shoes last only 3-5 months or 300-500 miles (limited durability). These issues lead to increased injury risk and suboptimal training.

User and Purchaser

Our target user is recreational runners in the United States who run between 10 and 40 miles per week. According to a 2017 survey (Appendix A) [14], 76% of runners fall in this category and due to their mileage, they need to frequently rotate and replace their running shoes. When selling our product, we would have several options. One would be to follow the B2C (Business to Consumer) model and deliver our shoe directly to runners. We could also follow the B2B (Business to Business) model and sell our product to rural outdoor equipment stores or sporting goods stores. Casual run clubs with members that fall into our user description are another potential purchaser. When testing and gathering feedback on our prototypes, we hope to work with runners at Dartmouth and in the surrounding area. We will reach out to members of the Dartmouth Cross Country and Dartmouth Track and Field DI Varsity teams, the Dartmouth Running and Dartmouth Triathlon club teams, and the Upper Valley Running Club.

State of the Art

Current Companies

The current shoe market is saturated with an array of companies that offer shoes with trade-offs primarily between performance and cost. While Nike, Asics, and New Balance offer many cost-effective options for singular trainers, brands like ON, Hoka, and Tracksmith offer premium products catered to optimizing performance and longevity. However, two companies have begun exploring the idea of a modular shoe: New Normal and Speedland.

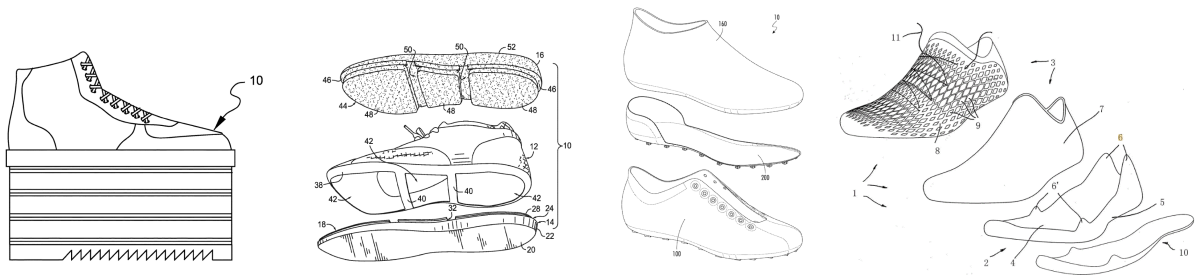
Both companies target the more niche market of trail running, but both have been designing shoes with removable components. New Normal has created a product called “Kboix” that has an extra-thick, swappable midsole. Speedland has released a design that also has an swappable midsole, but contains a removable carbon fiber plate. The shortcomings of both are a non-interchangeable outsole, a lack of replacement parts offered, and a high price tag. Since the shoes are designed to maximize performance, the interchangeability is purely for versatility reasons, resulting in prices ranging from \$190 to \$275.

Lastly, ON Running has explored an alternative to a modular shoe relevant to our focus on affordability. Deemed “CycleON,” the company has created a running shoe that is 100% recyclable and is obtained through a subscription. As the shoe’s lifespan comes to an end, the worn shoe is exchanged for a fresh pair, as soon as 90 days after being acquired [19]. No part of the design is removable, but the idea of recyclability could be paired with modularity to offer shoes at a lower price point. CycleON costs \$29.99 per month, or approximately \$360 per year, which is a cost-effective option for runners going through more than 3 pairs a year [19]. Where CycleON falls short is performance. The use of bio-based pebax foam and biodegradable structural components compromise the shoe’s energy return [20]. Furthermore, the versatility of the shoe is minimal as the tread is designed for pure road running [20].

Relevant Patents

Current modular shoe patents do exist, but aren't specified to the sport of running. Moreover, the designs devised to lock various upper, insole, midsole, and outsole components together aren't intuitive and easy-to-use. Rather, the existing patents use various methods of joining pieces together for aesthetic reasons or manufacturing purposes. US 2020/0113275 A1 is specific to platform shoes and provides a means to adjust the elevation of the heel [15]. US 11311072 B2, by Nike, describes a way to layer the upper, midsole, and outsole in a press-fit manner; the claim focuses on a gasket and tab-based locking mechanism, citing aesthetic customization as a benefit [16]. US 2020/0113275 A1, also by Nike, outlines a similar method of layering each shoe component. This time though, the layers are joined by a plurality of small, expandable plugs that extend into "traction elements" of the outsole [17]. Here, the design is meant to integrate modularity into cleats used by field sports, where traction on grass is of paramount importance. CN108135317B, filed by Puma, takes a different approach to keeping the upper and sole together: it uses a series of "sock-like" nesting structures to support the foot and hold components together [18]. The design may be modular, but Puma's focus was on creating a shoe that hugs the foot more effectively via a more fitted upper tensioning system. The form-fitting upper was created with the intent of increasing shoe grip and not necessarily for replacement purposes.

Existing patents ultimately explore the various means by which shoe uppers and sole layers can be joined together, but don't claim an intuitive and effective locking mechanism for shoes in a running context—particularly for the sake of durability and versatility.



Figures 1-4: Patents US 2020/0113275 A, US 11311072 B2, US 2020/0113275 A1, CN108135317B

Specifications and Problem-Solving Method

Alternatives

Given that our solution would seek to alleviate runners from the shoe rotation system, our team chose to utilize five general specifications that follow the general standards established by other footwear companies, in addition to requiring that our solution be safe, legal, and ethical.

Specification	Justification	Metrics (Units)	Margins
Safe/Legal/Ethical	To not incur further injury	Binary (Y/N)	Y
Affordable	To make running cheaper and more	Cost (\$)	< 200

	accessible		
Versatile	To optimize running on multiple surfaces	Hydrostatic Test (mm) Translation Force (friction coefficient) Stack Height (mm)	> 16,000 0.7-0.9 < 20
Durable	To ensure longevity and lower maintenance	Tensile Strength (mPa) Tear Resistance (N/mm) Lifetime (Mi) Stiffness (Shore Test A)	> 12 > 15 300-500 60-80
High Performing	To enable runners to perform at their best	Injury Percentage (%) Comfort Scale (1-5 Likert) Weight (oz)	< 35 > 4 < 11
Stylish	To boost confidence and motivation in runners	Rating (1-5 Likert)	> 3.5

Table 1: Solution Specification Matrix

Basic justifications for each specification are included (Table 1), but further discussion of each specification and metric can be found in Appendix B. After developing our specifications, we brainstormed many potential solutions (Appendix C) and ultimately settled on the idea of a modular shoe after comparing our ideas with an alternatives matrix (Appendix D).

Locking Mechanism Specifications

Specification	Metric	Margin	How We Tested
Affordability	Cost (\$)	< 200, when incorporated with shoe	Cost Estimate
Durability	Shear Strength (kgf) Hardness (Shore hardness A) Lock Strength (friction force)	> 24 60-85 Static Friction > Lateral Force	Instron Test Material Property Mathematical Analysis
Performance	Economy (%)	≥ 98	Mathematical Analysis
Ease of Use	Likert Scale (1-10)	≥ 7	User Testing
Safe/Legal/Ethical	Binary (Yes/No)	Yes	Consideration/Research

Table 2: Locking Mechanism Specification Matrix

Above is a table of specifications for our locking mechanism. The specifications we selected are what we deemed to be the most important qualities of an effective product for our targeted users.

The three specifications that we are prioritizing are the safety, legality, and ethics of our potential solution. Our solution must not endanger its users, violate current laws and policies, or create ethical issues in its manufacturing or use. We plan on evaluating the safety of our product by investigating injury

risk alongside any design concerns that may pose bodily harm to the user [21, 22, 23]. In measuring legality, we will evaluate our potential solutions by analyzing current policies and regulations within the United States regarding footwear/device manufacturing, wearables use, and other factors necessary to designing a general footwear solution [21, 22, 23]. The ethics of a given solution will be measured through a consideration of its holistic effects; we will evaluate the overall social implications of the product, whether it will endanger or exploit anyone who interacts with it (both consumers and producers), and whether it violates any laws regarding production or use [24].

Outside of these three specifications, we will consider four broader specifications, the first of which is affordability. Given that most runners within our user demographic seek to buy shoes that cost less than \$200, we are aiming to create a solution that falls within this range to ensure that our target users and purchasers would be willing to buy our product [25, 26]. We seek to reduce the cost of our solution as much as possible to not only make running cheaper, but also make it more accessible, particularly to runners who are younger, live in rural areas, or are of lower socioeconomic status. We will calculate the cost of our product by calculating an estimate based on the costs of individual parts and their sum.

Durability is another highly important specification; our idea of a modular shoe will not be successful if the locking mechanism does not last. Increased durability decreases the resources necessary to maintain strong performance and is an important selling point for our product. Peak shear force on a shoe ranges from 87 N to 235 N, or from 8.87 kgf to 23.96 kgf [27]. Therefore, we expect our locking mechanism to have a shear force capacity that is greater than 24 kgf. We tested the shear strength of each mechanism by using the Instron machine. The hardness of the locking mechanism can be measured through the Shore A Test. Shore A Test values of 60-85 are ideal, as it would allow for enough flexibility while still being sturdy [28]. We have selected to 3D print our locking mechanisms in TPU filament which has a hardness of 85A, so is both flexible and sturdy enough. We quantified our locking strength through the general frictional force as that would be the main force that contributes to locking the mechanism in place and resisting shear force that would slide and unlock the mechanism. Coupled with the Instron testing that showed how much force the locking mechanism could resist if pushed in a perpendicular direction, both these specifications allow for measuring how well the lock is able to stay secure under stress.

Although we want our product to perform very well, performance is not the ultimate priority for our target market. It is still an important specification, however, as the shoe should allow runners to make progress towards their goals and reach their running potential. Running economy is a means by which we can measure the effect of our locking mechanism on overall shoe performance. By mathematical analysis, we will calculate the maximum weight that the locking mechanism can weigh while still keeping the economy at greater than 98% [29].

Our final specification is ease of use. The locking mechanism we create should be intuitive and easy to use. We will run user tests and review of our locking mechanism and collect Likert scale ratings. We will aim for our locking mechanisms to receive a positive average Likert score greater than 7 out of 10. Additionally, in our user testing, we will collect comments from both a “think aloud” test and post-testing reflections. This qualitative feedback will also help us understand how intuitive the locking mechanisms are.

Prototyping

Sketching and Ideation

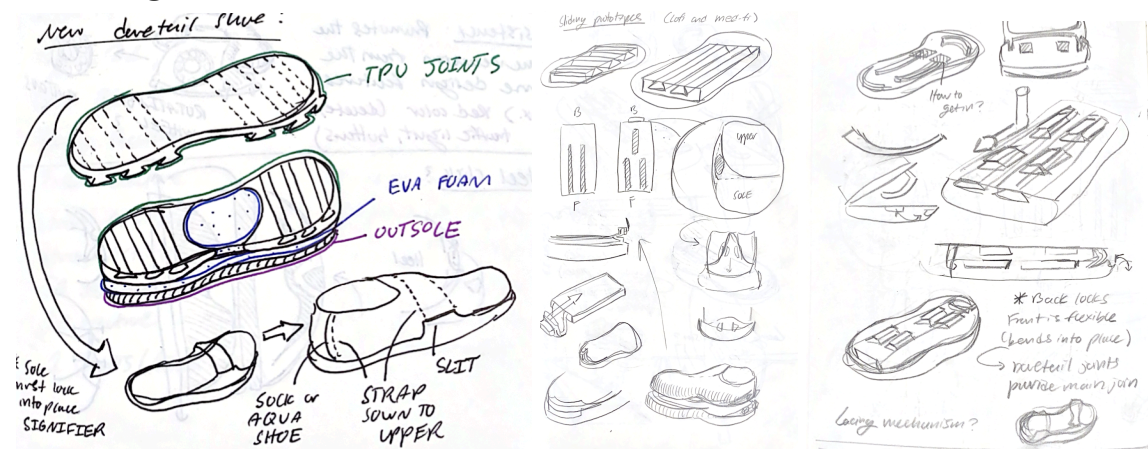


Figure 4: Sole and Upper Attachment Method Brainstorming

We began by breaking down the complex geometry of a shoe into planes that could potentially hold our locking mechanism; we decided to take a planar approach along the footbed, since it would facilitate an easier removal of the midsole and outsole. Our planar simplification allowed us to focus on the unique part of our solution because conventional shoe design is already well established. From there, we brainstormed different methods of attaching the upper and sole planes together. Our early ideas drew both upon existing mechanical mechanisms and our own creativity. It was after we had a few potential ideas that we imagined how to integrate each mechanism within a shoe, and from there, we selected the most promising designs.

Initial Locking Mechanisms

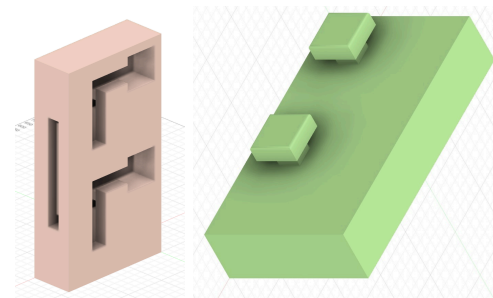


Figure 5: Initial Channel Mechanism

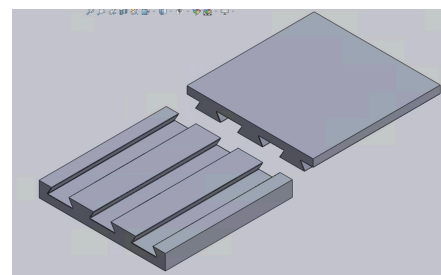


Figure 6: Initial Dovetail Mechanism

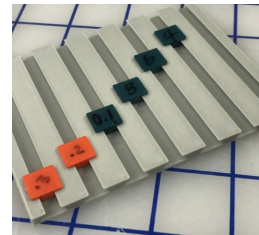
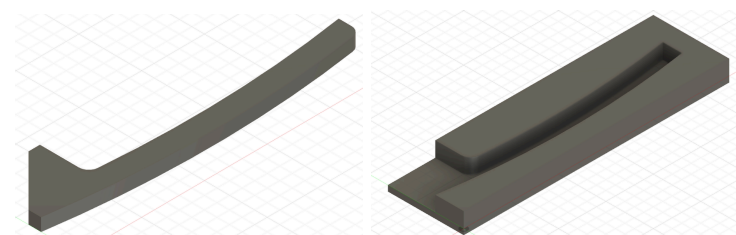


Figure 7: Sliding Arc Mechanism**Figure 8: Tolerance Testing**

We developed three initial locking mechanisms after brainstorming different means by which to connect an upper and sole. The first was the channel mechanism (Figure 5). In this model, two pegs enter two channels and slide through them until they lock. The second was the dovetail mechanism (Figure 6). In this model, flared connectors from one piece slide into place within the other piece. Finally, we devised a sliding arc mechanism (Figure 7). However, after further consideration, we found that this mechanism did not lend itself well to shoe integration and chose to eliminate the idea.

From our 3D printing of these initial locking mechanism models, we found that it would be a challenge to find the appropriate tolerances. We did not want the dovetails, for example, to be so tight that it was impossible to slide them together or pull them apart, but also could not let them be too loose so as to not effectively lock. In order to select an effective tolerance we printed small pieces of varying different sizes (Figure 8). We tested tolerances ranging from 0.3mm to 0.04mm and used the feel of those tests to determine what tolerances to use for our larger test/prototype prints.

Foot Geometries

**Figure 9: Shoe Last and Molded Foot****Figure 10: Scanned Foot**

Though we elected to focus primarily on the locking mechanism by which we would connect an upper and a sole, it was key to still consider how the mechanism would function with a foot and in a shoe. We made the decision to model our work on a men's size 9 foot. We first 3D printed a shoe last scaled to the appropriate size, then also molded a real foot of the correct size. We later used these foot models to build the shoe around. We also taped the molded foot to smooth it out, then scanned it into Solidworks. This allowed us to have the complex geometry of a foot available digitally and also have an accurate sole shape on which to model our locking mechanisms.

Rudimentary “Works-Like”



Figure 11: Initial Implementation of Channel Mechanism

Our initial attempt to incorporate the locking mechanism into real footwear used the channel design. The channels were added to a sole outline for appropriate integration into a shoe and we 3D-printed the components in PLA. We modified an EVA sandal, inserted our locking mechanism, and attached a pre-constructed synthetic upper using interface fabric and hot glue. The model foot fit in the shoe, but the stiff and thick nature of the locking mechanism meant there was poor sole flexibility. Furthermore, the channel mechanism lacked friction at each insertion point, and thus the upper to sole connection was unstable.

Final Solidworks

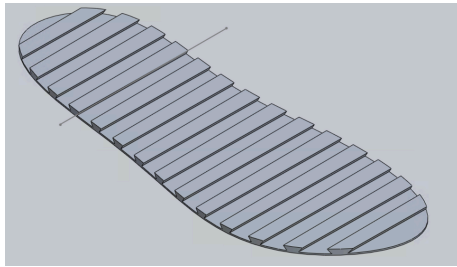


Figure 12: Final Dovetail Mechanism

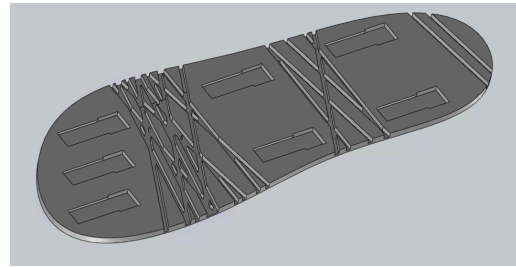


Figure 13: Final Channel Mechanism (Female Side)

After testing many different tolerances, we printed our final dovetail mechanism with a tolerance of 0.15mm and our final channel mechanism with a tolerance of 0.015 in (~0.38mm). We modeled the dovetails and channels on the scanned shape of the sole and printed them in TPU. Both received slight modifications based on information we had learned from previous iterations. They were both very thin and accordingly quite flexible: their base was only 1 mm thick. We tapered the dovetails slightly to create a one-way entrance and more secure lock (Figure 12). We included hinge points in the channel mechanism for greater footbed flexibility at key areas of foot flexion (Figure 13).

Final Prototype



Figure 14: Tape Upper Pattern



Figure 15: Final Prototype

In order to make our final prototype, we wanted to show proof of our broader concept by putting together a complete shoe. We taped our molded foot to create the pattern for a shoe upper and then cut out pieces of fabric that were traced out of the taped pattern, with some extra overhang on the parts touching the flat part of the sole so that they could be attached properly. The mesh layer of the upper was created out of four layers of fabric. On the inside and outside layer, a fine mesh fabric was used to enhance breathability while also maintaining structural integrity. Sandwiched between these fine mesh layers are two more open mesh layers, which are visible under the fine mesh layer in figure 15. After attaching all these layers together with glue at their corners, fabric reinforcement (black fabric in figure 15) was used where the eyelets were placed to ensure durability. Six eyelets were attached for each lacing side. After these steps were complete, the fabric mesh was put around the shoe last, and darts were cut around the overhanging (extra) parts of the four-layer mesh fabric so that they could be shaped to the shoe last and glued to a flat sole piece shaped like the top taped piece in figure 14.

This three dimensional mesh with a last bottom was then glued to a modified black foam Birkenstock shoe with its straps cut out to fit with the mesh and last layer. The foam Birkenstock had somewhat raised edges on its sides, allowing for a better connection between the white mesh layer and forming a continuous “bite line.” Since the shoe pattern was cut directly down the side of the heel, the heel needed to be connected with a heel counter on the inside of the shoe, which also gave the shoe mesh more support in being able to stand up on its own while also providing greater comfort and durability. Black fabric reinforcement was again used to patch up the seam on the outside of the heel. Laces were taken from a used HOKA shoe.

Both locking mechanisms were composed of two planar sides that would be joined together. As such, one piece of the locking mechanism would be attached to the sole, and the other part that joins with it would be connected to the sole. In the final shoe, the channel locking mechanism was used, with the male side of the locking mechanism being glued onto the upper.

The sole was composed of the female side of the locking mechanism, a layer of EVA foam, and then a tread pattern at the very bottom that were all glued together. These two pieces, the sole and the upper, could then be attached and secured via the interface between the male and female channel mechanism. Although this will be further elaborated in the user testing section, further design changes will need to be implemented in order to make the channel mechanism more simplistic and provide more feedback to the user to make sure they can understand when the locking mechanism is fully locked. Even so, the final prototype comprises a modular upper and sole that is relatively simple to connect and is a good representation of what the final product would look like.

Testing and Analysis

Friction Analysis

The main locking force of our mechanisms is a mixture of a clamping force applied by the locks and a frictional force that would resist any movement of the locking mechanism during running. From previous studies [30], it has been established that there is a max ground reactive force (the maximum amount of energy put into the ground) of around 2-3 body weights, with a lateral peak force that is merely the vertical component of this force and is at most 0.5 body weights. Static friction will be the force that resists this motion, and thus a simplified way to measure the locking force would be to measure if the

static friction between the two locking mechanisms is enough to prevent movement. To do this calculation, the simple formula for static friction is used:

$$\text{Static friction} = \text{coefficient of friction} \times \text{normal force}$$

The goal of this calculation is to demonstrate if the static friction is significantly greater than the peak ground reaction force. TPU with a hardness of 85A has a coefficient of friction at about 1.0 [31], so plugging in values into the formula yields:

$$\text{Coefficient of friction} \times \text{max ground reactive force} = 1 \times 2 \text{ BW}$$

Even with a low estimate for the ground reactive force, the overall static friction is significantly greater than the maximum lateral peak force ($2 \text{ BW} > 0.5 \text{ BW}$), meaning that the locking mechanism will be secure during running. However, this is a very simplified version that dismisses complex foot geometries and frictional forces in the ground. Further simulations and tests regarding frictional analysis would be required and the orientation of the channels and dovetails would need to be further optimized so that unwanted unlocking of the mechanisms does not occur during running.

Footwear Economy Analysis

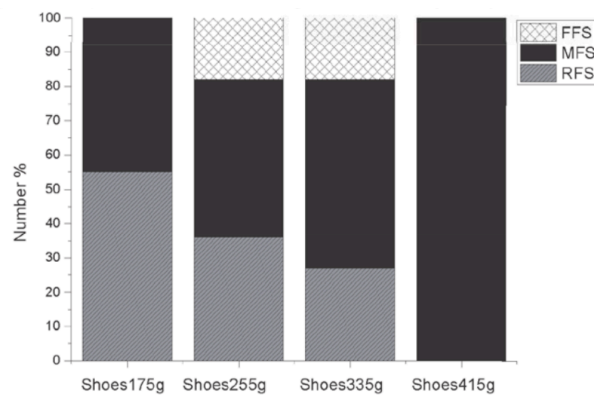


Figure 16: Different masses of shoes with strike index [32]

An average running shoe weight ranges 200 to 300 grams and an additional 100g of weight only causes a 1% decrease in performance [29]. However, added mass may change a runner's gait pattern, increasing risk of injury as they move from a rear foot strike (RFS) to a mid foot strike (MFS). In order to prevent injury, we set the specification that our locking mechanism should not weigh more than 125g. Both of our mechanisms passed this test: the dovetail mechanism weighs 104.6g and the channel mechanism weighs 89.9g.

Solidworks Finite Element Analysis Simulation

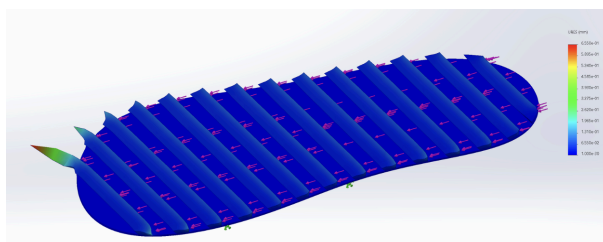


Figure 17: Dovetail Mechanism FEA

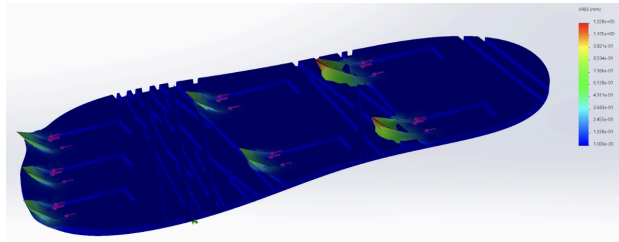


Figure 18: Channel Mechanism FEA

After modeling our final locking mechanisms, we ran them through a Finite Element Analysis (FEA) simulation in which we examined the displacement of mechanism material when 25 kg of force was applied in the forward-backward direction. We elected to test the effects of 25 kgf because it is above the maximum shear force that a shoe experiences [27]. Under the specified force, the dovetail mechanism would suffer a max material displacement of 0.65 mm at only the very tip of the front joint (Figure 17). Under the same conditions, the channel mechanism would sustain a maximum displacement of 1.2 mm at only the very front edge of the rear channels (Figure 18). These values of displacement are relatively small and reflect the effect of sustained maximum force; therefore, they reflect the sturdy geometry of our locking mechanisms. In addition, it is notable that we found TPU as a material to be remarkably good at rebounding from deformation; when experiencing sustained pressure of more than 30 kg from the instron machine in our shear tests, the material returned to its original shape when the force was removed.

Instron Shear Test

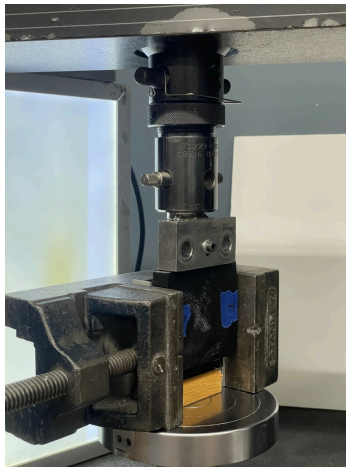


Figure 19: Instron Shear Test Setup

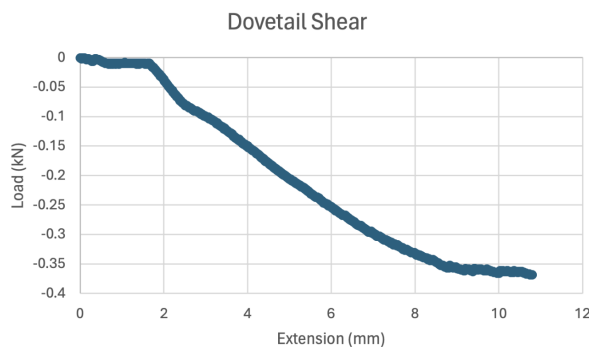


Figure 20: Dovetail Shear Test Data

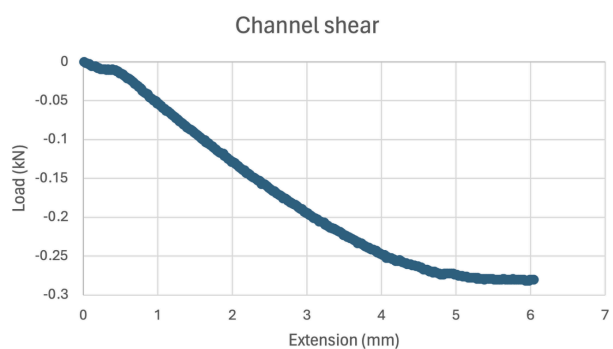


Figure 21: Channel Shear Test Data

To conduct the shear force capacity test with the instron machine, we needed to simplify the geometries of the locking mechanisms from sole shaped to rectangular. We locked the mechanism together, then set up the instron machine to push on only one side of the clamped mechanism in the direction perpendicular to the locking direction (Figure 19). Because peak shear force on a shoe can reach

23.96 kgf, for the test to be successful, we expected our locking mechanisms to be able to sustain shear forces greater than 24 kgf [27]. The dovetail mechanism managed a maximum force of 368.73 N, or 37.6 kgf (Figure 20). The channel mechanism managed a maximum force of 282.4315 N, or 28.8 kgf (Figure 21). Therefore, both mechanisms met our benchmark for shear force capacity.

Instron Insertion Force Test

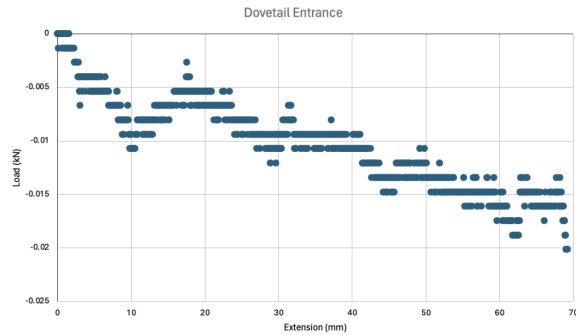


Figure 22: Dovetail Insertion Test Data

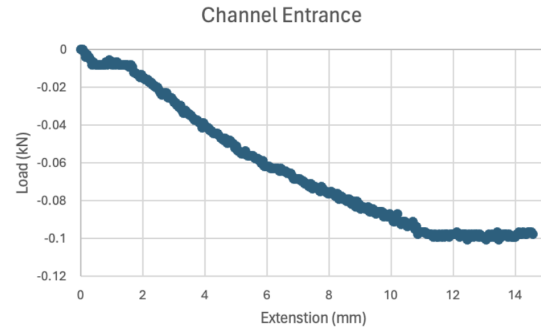


Figure 23: Channel Insertion Test Data

We also tested how much force it would require to insert the different locking mechanisms using the instron machine. We continued with the rectangular geometry locking mechanism pieces and held the base side lightly in place by hand so that the clamping force from the sides would not disturb the entrance of the mechanism. As our benchmark, we found that the average untrained person can exert a force of up to 10.2 kgf when pushing or gripping objects with their hands [33]. The maximum force required by the instron to insert the dovetail mechanism was 20.10363 N, or 2.05 kgf (Figure 22). The horizontal bars in the dovetail graph is a visualization of the applied force by the machine staying the same for multiple extensions. The maximum force required by the instron to insert the channel mechanism 97.96843, or 9.99 kgf (Figure 23). Both locking mechanisms were inserted with less than 10.2 kgf and therefore passed the instron insertion force test.

User Testing and Feedback

A sample of recreational student runners ($n = 24$) were asked to join the sole and upper of the final prototype that utilized the channel mechanism. Users were sourced predominantly from the Dartmouth Triathlon Team, and the minimum qualification was >3 runs per week. No prior knowledge was given regarding how the locking mechanism works, and no instructions were given during the test. The success rate and time to lock of users were recorded.

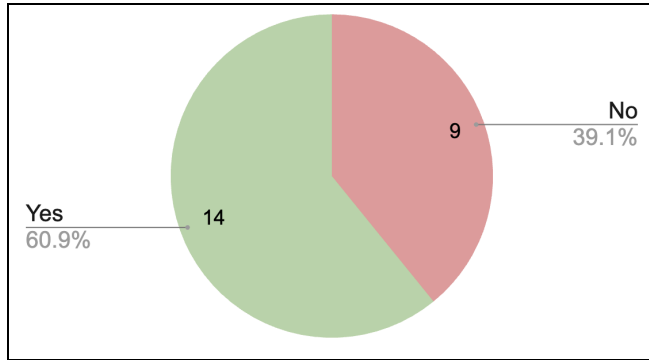


Figure 24: User Success of Locking Mechanism

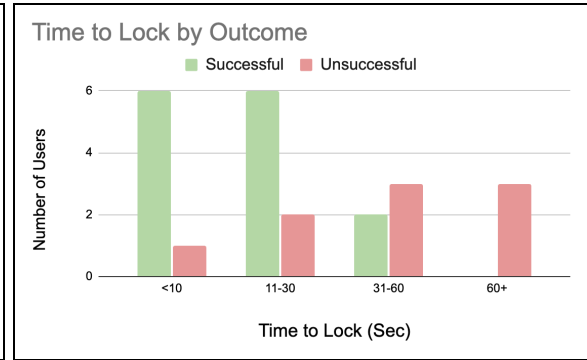


Figure 25: User Time to Lock by Success

A significant outcome of our user testing was that the majority of users (60.9%) successfully joined the upper and sole. More importantly, however, the lack of success among some users revealed areas of our design to improve upon. >50% of successful locks took < 30 seconds or less, while >65% of unsuccessful locks took more than > 30 seconds. This showed that successful users were faster at attaching the components together, while unsuccessful users were significantly slower. This may appear as an intuitive, obvious observation, but upon receiving qualitative feedback, it's actually indicative of the deeper reason for lack of success. Users cited their experience as a "learning process" in which attaching each piece became immediately clear upon realizing how they fit together. It was after the first attachment process that users became more adept at joining the two together; many tried again and succeeded more quickly than before. Thus, the reason why the initially successful users were faster was because they examined the parts to understand how they fit together and immediately knew what to do. This "learning process" was further observed as part of the time-to-lock test when users were asked to think aloud.

"Looks like a puzzle," "I see one part is the bottom, and one is the top," "That was easy"

- Successful users

"I'm confused on how the pieces go together," "Does it slide?," "I can't line up all the pieces"

- Unsuccessful users

We had two main takeaways from our user testing: signifiers could be added to convey how the mechanism worked and a confirmation of total lock could be added to ensure proper user. Signifiers would reduce the time duration of the "learning process" that users cited and shorten the total attachment time. A confirmation of lock would help users know when they've succeeded and prevent the shoe from being used without being properly attached. Both could be accomplished using visual, graphic elements that could include colors, arrows, and other indicator signs [Appendix E].

Ethics and Sustainability

The original idea of the modular shoe concept centers around being able to have interchangeable soles that would be cheaper and more versatile than having one or multiple shoes. Given this premise, the ability to easily change shoe soles given the terrain and seamlessly swap out worn out soles for new ones means that the running process itself would be more optimized to promote efficacy and safety, as proper

soles would be used. In the realm of sustainability, this product will help to reduce waste in discarded but relatively pristine uppers, meaning that both the sole and upper could both be used to their fullest capacity. This, in turn, also reduces the costs of having to buy entirely new shoes when only the soles are worn out, as individual soles would cost much less overall than having to buy both an upper and a sole (a new shoe).

Despite these overall sustainability and ethical solutions built directly into the modular shoe design, there still remains issues in the harvesting and production of materials that are used in the current design of the shoe. EVA foam [34] and TPU [35] production is harmful, producing toxic chemicals that are harmful for workers and the environment. Although the following of rules and regulations reduces the harm that producing these materials incurs, evaluation of novel materials that do not have the same harmful production effects would be effective. For example, EVA foam could be substituted with talalay latex, which is a natural alternative to EVA foam that has no harmful effects in production. Bio-based TPU is also a substitute to the synthetic TPU that could be evaluated in reducing the harmful off-gassing that results from TPU production.

Evaluating various other locking mechanisms is also a top priority as the locking mechanism should ideally never fail and be able to endure a plethora of environments. Current testing is still rudimentary and assumes relatively planar geometries, and thus more testing would be necessary to ensure the safety of the current designs alongside further optimization.

More research will need to occur in the sourcing and manufacturing parts of the modular shoe. The sourcing of raw materials required for alternatives to EVA foam and TPU would be necessary to further scrutinize the proper ethical treatment of workers in both its harvesting and production. This includes evaluating labor practices and other ethical aspects related to harvesting. On the manufacturing side, similar research will need to be done to evaluate the labor practices at the manufacturers, making sure that there is proper local employment and alignment with proper labor policy in the given region (most likely China or Taiwan due to existing shoe-making infrastructure).

A final aspect of the product that is in consideration is simplifying parts so that the amount of materials needed could be reduced. For example, our group evaluated the possibility of creating a sole solely out of TPU that would have an infill to mimic the properties of the current EVA foam in the sole part of the shoe. In doing so, the manufacturing and sourcing problems would become diminished and the shoe would also be much easier to assemble and modify.

Business Plan

Business Viability

The central reasons why a business leveraging our product would succeed would be our unique value proposition and business model. A shoe with interchangeable soles would offer the benefits of a premium shoe rotation at a lower cost. While conventional running shoes cost approximately \$390 annually (3 x \$130), our modular shoe system would cost \$220 (1 upper = \$70, 3 soles = 3 x \$50) [36]. This significantly lower cost for training benefits and injury prevention would appeal to our target user group of recreational runners. Furthermore, our business model of selling both upper and sole packages along with individual soles would yield consistent revenue since buyers would purchase a complete shoe and then continually repurchase soles. Once users have the upper and a few soles, we could expect low churn because the barrier to purchasing an entirely new shoe from another brand would be higher than

simply buying a new sole. As such, our business model would have multiple streams of revenue coming from complete shoes, individual uppers, and individual soles

Business Implementation

To begin our business venture, our initial focus would be on investing heavily in prototype development. Our physical product, the modular shoe, will determine our value proposition and ultimately drive sales. Therefore, we would dedicate significant time and resources to ensuring that our locking mechanism meets all specified requirements and can be seamlessly integrated into a high-performance shoe. This phase will draw upon our existing skill sets, potentially requiring us to expand our range of prototyping skills, and may later involve outsourcing labor to professionals capable of translating prototypes into production-ready products. Once we reach a stage where the product is ready for mass production, we would consider filing a provisional patent application, followed by a non-provisional application.

In terms of marketing our product, we would aim to target consumers who already engage with similar companies such as Hoka, On Running, Salomon, and Brooks. Given our understanding of the recreational running market (12 million American runners) and the pricing range of these companies, we anticipate that once runners grasp our unique value proposition, they could become regular customers. To entice runners to switch from other brands, we would offer competitive pricing on whole shoe packages and generate revenue by selling individual soles at a slightly higher margin. We plan to reach these customers through various channels, including race expositions and social media platforms such as TikTok, Instagram, and YouTube. As outlined in the user and purchaser section, we will initially engage with local organizations such as Dartmouth Club Sports, Dartmouth D1 Athletics, and the Upper Valley Run Club. Additionally, we may explore the option of providing free pairs to professional or celebrity runners with a significant online presence.

Regarding distribution, we will adopt a Business-to-Consumer (B2C) model initially, utilizing an e-commerce platform like Shopify and partnering with a reliable shipping company such as FedEx for inventory delivery. Future growth opportunities may involve expanding to a Business-to-Business (B2B) approach, wherein we would collaborate with independent running shoe stores for resale, utilizing comparable shipping arrangements.

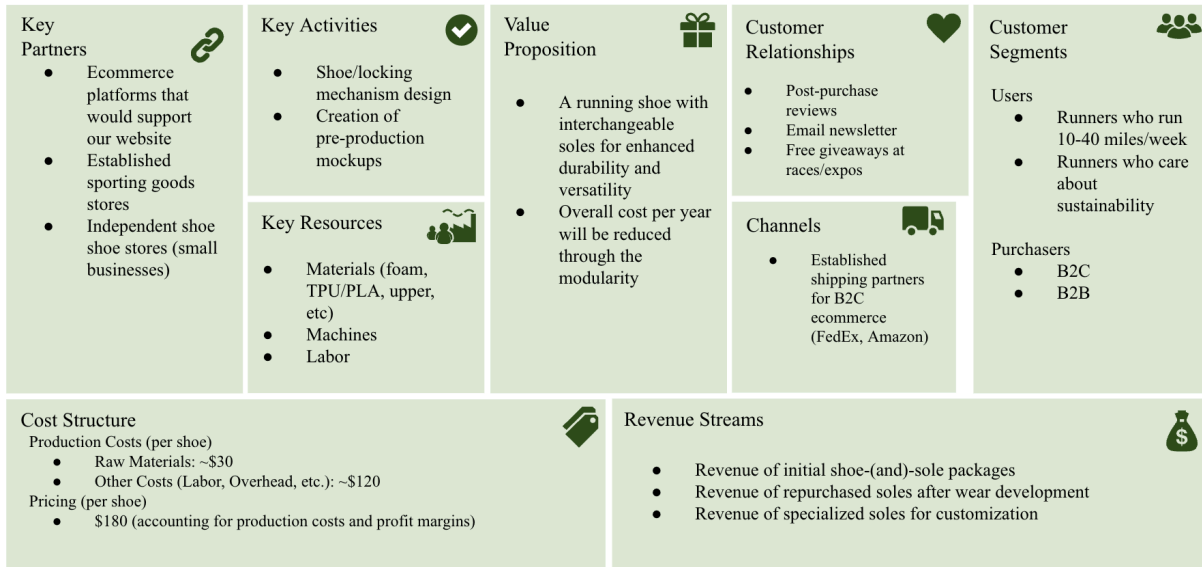


Figure 26: Lean Canvas of Business Plan

Reflections

In the process of prototyping and turning our concept of the modular shoe into a physical and usable model, our team learned a lot about specific concepts in shoe making alongside broader aspects of engineering. In terms of shoe making, our team effectively learned how shoes are made from the bottom up, with team members learning sewing techniques, patterning, and making a last (among other specific skills in shoemaking). This was paired with a range of technical skills that were taught during the T&T sessions that helped with the overall building portion of our prototypes and finite element analysis of the locking mechanism. In regards to the overall engineering and design process, our team became much better at problem-solving over the course of the term, whether it was actual design problems, issues in communication, or failure of an entire prototype. Mitigating these failures was primarily resolved through constant iteration, as failures became new areas for growth that allowed us to develop a final product that is significantly advanced from our initial prototypes. This iterative process was the greatest moment of learning that echoed not only engineering thinking, but the necessity for collaboration, communication, and innovation. The more human side of engineering was exposed to us, especially during the user research and testing, as we were able to truly visualize and quantify a problem that was described by our potential users, and this was a rewarding experience as we not only were able to learn from our users, but have our prototype truly field tested with novel eyes.

Thayer research resources were also a big boon for our team, as we learned a lot about the research and patenting side of things. This was a part of the larger help we received from the amazing staff at Thayer, and we came to rely on the expertise of Lee, Cody, Mingwei, and Professor Murnane in guiding us through the process of turning our concept into a reality. As a whole, our interactions with faculty/staff and internal interactions created a greater appreciation and understanding about overall teamwork and mentorship, which was a learning experience we all had on Team 6. Ultimately the team members are the foundation of a project, and ensuring effective communication and trust in other's abilities allowed for our team to learn together and prosper.

However, within the engineering process, there was still some room for improvement. As a part of the iterative process, we had many failed 3D prints that we waited far too long for. This was primarily due to our lack of making test pieces that would test for tolerances or structure in general. Over time, our team learned to make test prints more often (this was a major learning part of the iterative process), but there still needed to be an openness to exploring and testing pieces rather than trying to make the final intended product on the first try. In the actual locking mechanisms, the feedback we received from potential users was very insightful and led us to really focus on trying to make the locking mechanism more intuitive and communicative. Although we saw moderate success on the channel lock, feedback showed that there needed to be more communication in the lock, through both auditory clicking that suggests a secure lock, and visual cues such as pairing insert and channel colors to allow for a more intuitive understanding of how the lock works. By changing our lock from a solid and silent combination to a product that gives more feedback, we hope to make our product more user friendly. A final overall point regarding the entire process of both design, prototyping, and analysis would be our need to be more strategic with our prototypes and their development. Referring back to the need to do test printing, our team needed to make many more prototypes, as each prototype was significantly better than the last. By only having a handful of prototypes made, we were not able to fully iterate as well as we would have hoped, and by setting more defined goals in our prototypes before we actually prototype, we believe that our team would be able to more quickly develop prototypes that also can be more informative.

Future Plans

Our primary short term goal would be to iterate on our current prototype and create a fully-functional model. The directionality of the channel locking mechanism would be altered to create a more secure fit and signifiers would be added to streamline the attachment process. We would also plan to brainstorm an additional, external locking component to further secure the channels; this could take on some form of pin, hook, or strap. With these modifications, we would conduct more extensive user-testing to investigate how repetitive use and instruction affects the ease of use.

Beyond the channel mechanism, we would also attempt to integrate the dovetail mechanism to determine its viability. Once the potential of both designs have been explored by fully implementing each in a shoe, we would move on to test both by having users wear our prototype. This next phase of user testing would bridge the gap between a functional locking mechanism and an actually functional shoe.

Our long-term goals would include brainstorming additional locking mechanisms, exploring the option of a rig for easy attachment, and applying locking mechanisms to more forms of footwear. Our current design focuses on joining the flat planes of the upper bottom and sole top, but a more integrated design could include an attachment to the arch banding of the upper. A rig could also simplify the design by removing certain user considerations in the shoe components, but may prove impractical in real user scenarios. Lastly, we would try expanding our mechanism in boots and casual shoes to offer adaptability in different weather conditions. While interchangeable soles could be an effective technology in a running context, they could also prove useful for users who experience swings in weather and consequently have a need for appropriate soles for a variety of outdoor environments.

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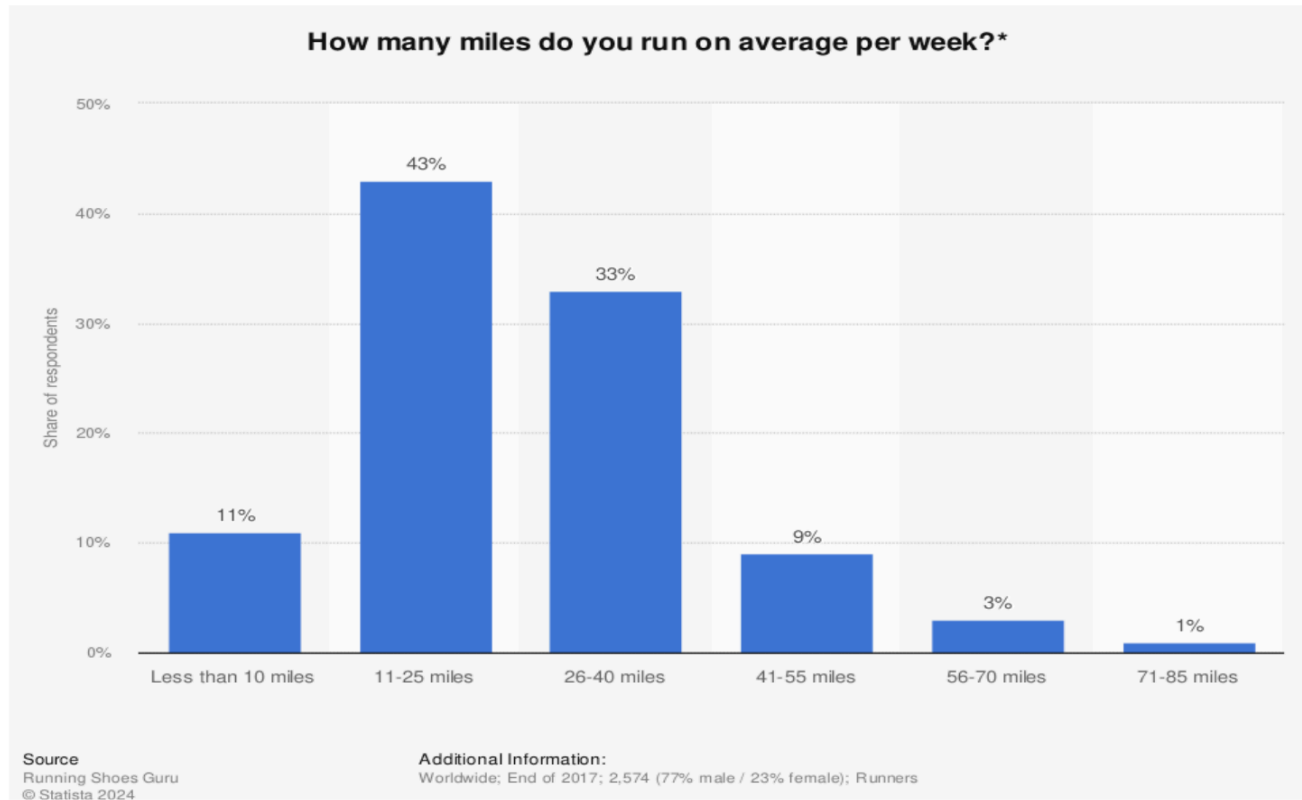
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Appendix

Appendix A: Runner Statistics [14]



Appendix B: Additional Information on Our Solution Specifications

The three specifications that we are prioritizing are the safety, legality, and ethics of our potential solution. Our solution must not endanger its users, violate current laws and policies, or create ethical issues in its manufacturing or use. We plan on evaluating the safety of our product by investigating injury risk alongside any design concerns that may pose bodily harm to the user [21, 22, 23]. In measuring legality, we will evaluate our potential solutions by analyzing current policies and regulations within the United States regarding footwear/device manufacturing, wearables use, and other factors necessary to designing a general footwear solution [21, 22, 23]. The ethics of a given solution will be measured through a consideration of its holistic effects; we will evaluate the overall social implications of the product, whether it will endanger or exploit anyone who interacts with it (both consumers and producers), and whether it violates any laws regarding production or use [24].

Outside of these three specifications, we will consider five broader specifications, the first of which is affordability. Given that most runners within our user demographic seek to buy shoes that cost less than \$200, we are aiming to create a solution that falls within this range to ensure that our target users and purchasers would be willing to buy our product [25, 26]. We seek to reduce the cost of our solution as much as possible to not only make running cheaper, but also make it more accessible, particularly to runners who live in rural areas or are of lower socioeconomic status.

Versatility is another important specification, as we seek to create a solution that is able to address the adaptability in run training. Thus, we seek to develop a product that allows runners to move effectively on multiple surfaces, whether on wet, jagged, or smooth terrain. Our solution would ideally help with waterproofing the shoe for wet environments; $>16,000$ mm on the hydrostatic test would be ideal to allow for robust water resistance [37]. Good grip on the sole of the shoe is also crucial for reducing slipping and improving performance. Therefore, developing a solution that makes the friction coefficient of the sole be around 0.7-0.9 would be necessary [38]. A stack height of less than 20 mm is preferable, as less material between the foot and the ground allows for better responsiveness to regular or irregular terrain. A significant amount of trail running shoes have stack heights less than 20 mm, while some popular shoes have one of less than 10 mm [39].

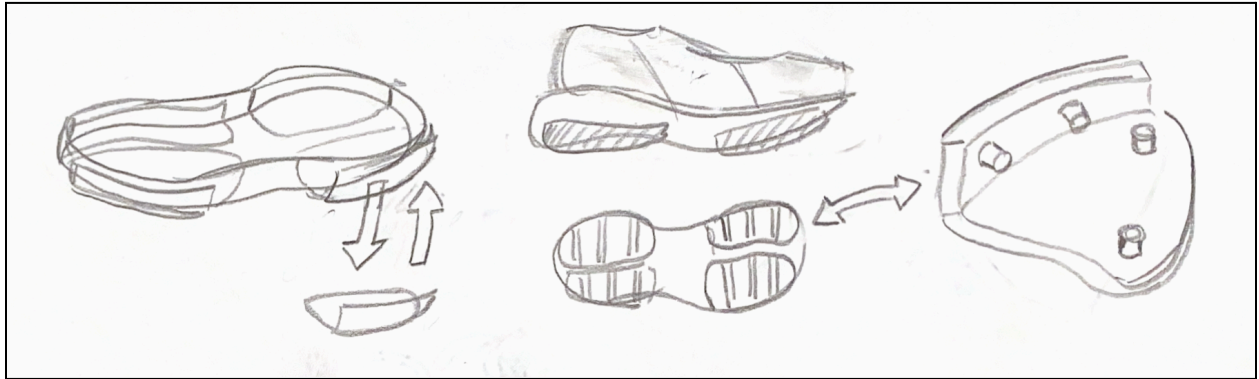
We want our solution to be durable; a product is not worth purchasing if it does not last. Increased durability will decrease the resources necessary to maintain strong performance and would not need to be replaced as often. Ideally, the solution should match the industry standards of >12 mPa in tensile strength and have a tear resistance of >15 N/mm [28]. The hardness (measured through the Shore A Test) would also be an important factor in the tread, as having less flexible treads allows for less tread wear. Shore A Test values of 60-80 are ideal, as it would allow for enough flexibility while still being sturdy [28].

Although we want our product to perform very well, performance is not the ultimate priority for our target market. It is still an important specification, however, as the shoe should allow runners to make progress towards their goals and reach their running potential. Injury risk is a key factor of performance, as minor overuse injuries can make training impossible. Technique plays an important role in reducing injury risk, but so do shoes. To test the injury mitigation capabilities of our shoe, we will have runners trial our potential solution; our goal rates of injury risk would ideally be 50% (standard yearly rate) or lower [9, 10]. Fit is another factor of performance because uncomfortable running shoes with narrow toe boxes inhibit proper knee flexion and reduce balance, which inherently leads to gait inefficiencies [40]. Other factors of performance include breathability and the general feel of a shoe; increased comfort improves running economy [41]. Thus, we will measure comfort through user testing that must result in an average likert scale value of greater than 4 out of 5. Weight is a final factor of consideration as greater

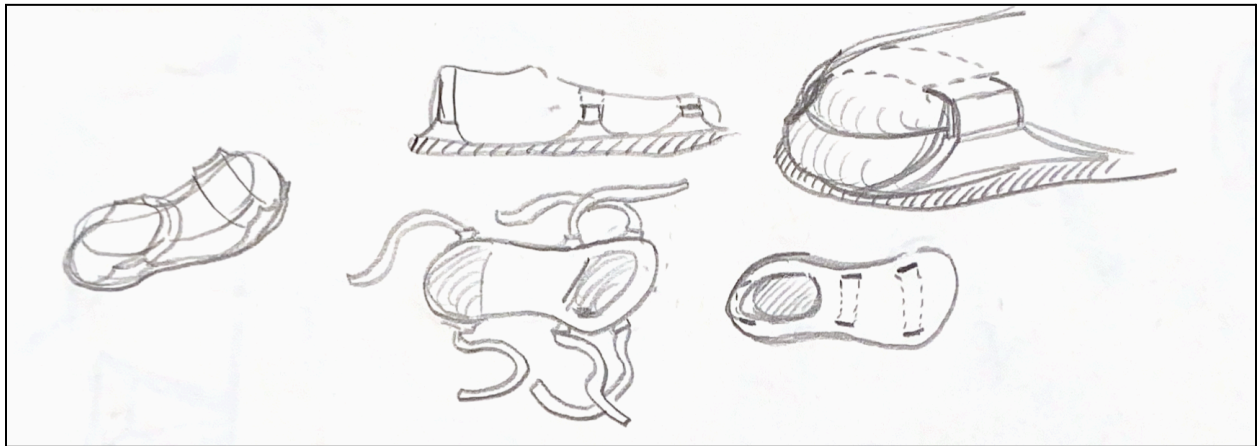
mass yields less energy efficient shoes. Accordingly, our ideal running solution should weigh less than 11 ounces. Attachments that may be part of our solution should not weigh more than 2 ounces as even a 100 grams increase is enough to noticeably affect performance [42].

Style is the final specification that we are considering, but we value it far less than the other specifications. It is still important, however, because a stylish solution can help runners to feel confident and more motivated to run [43].

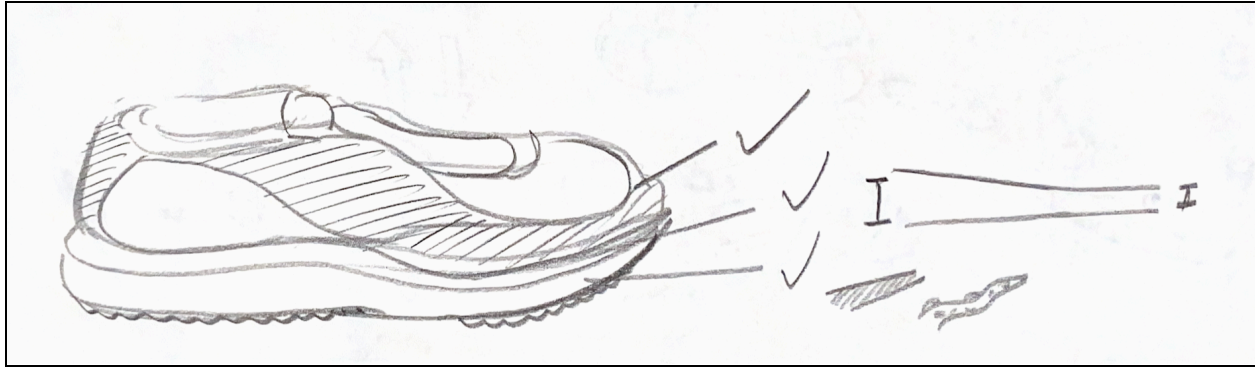
Appendix C: Brainstormed Alternative Solutions



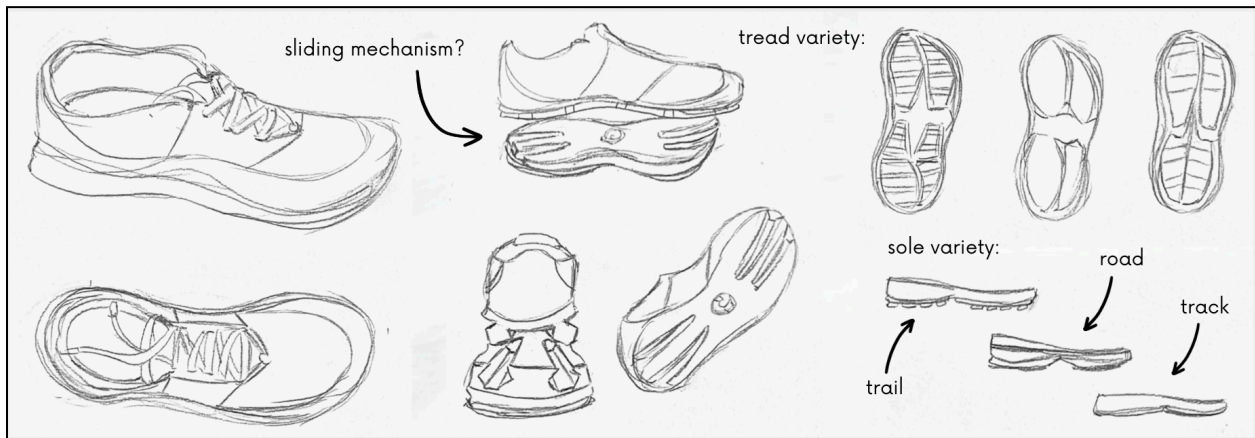
Sole Attachments: In order to get versatility out of a shoe, we first considered a model in which a single shoe could be designed to allow for sole attachments. These would have a low profile so as to not change the experience of running with the shoe, but could provide different treads and grip for different conditions. The primary spots of contact between shoe and ground are at the ball of the foot and at the heel, so those are the areas we would target when placing the attachments. These attachments would be easily removable and replaced and could ideally even be used on worn out shoes.



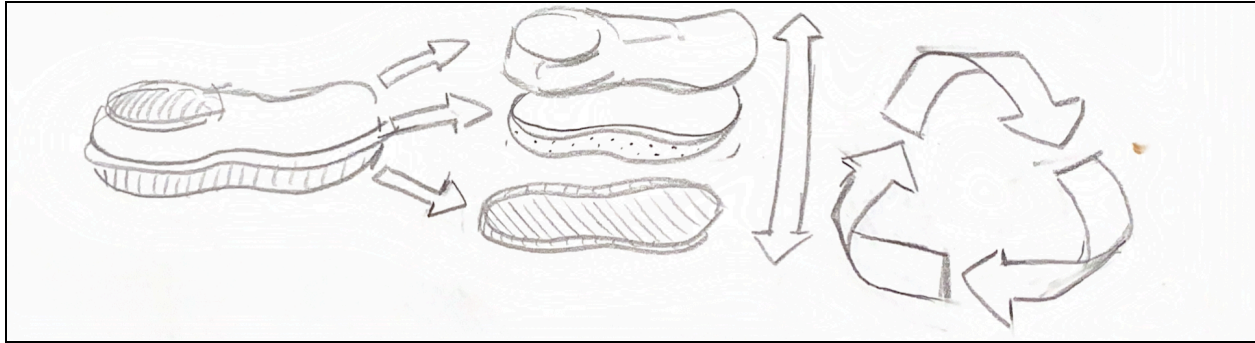
Barefoot Pads: Barefoot running has become increasingly popular on soft surfaces such as on treadmills or dirt roads. It has been shown that barefoot running has numerous benefits such as reducing injury risk and stress on various parts of the foot and ankles. However, there are issues when running in terrain that is hard or has sharp foreign objects [44, 45]. A barefoot pad could be a low profile, easily interchanged option for barefoot runners that addresses these issues. The material could be engineered to be anti-slip and grippy on different running surfaces.



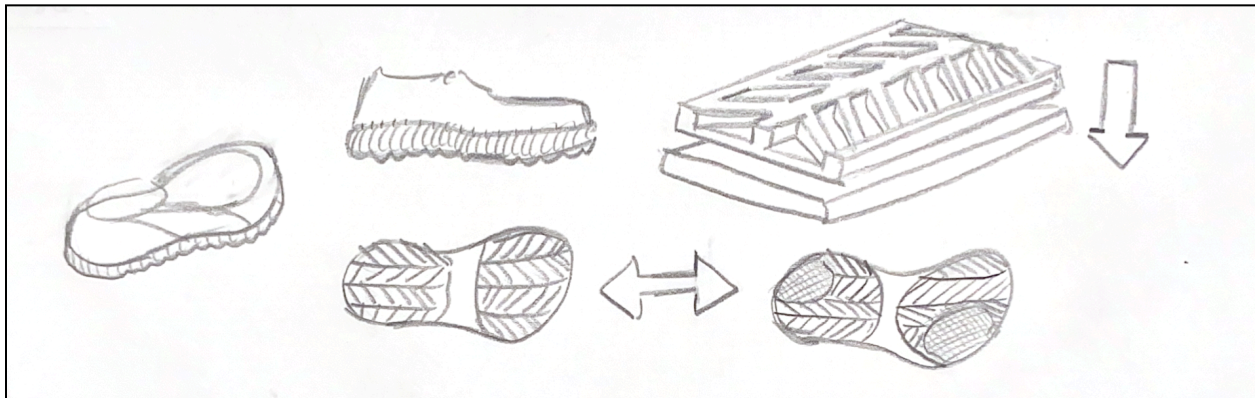
Hybrid Shoe: In a hybrid shoe, we would find a way for a singular pair of shoes to perform well on a wide variety of surfaces. While maintaining a high level of durability, we would want a versatile all-in-one model that could function as many different types of shoes. Programmable materials could be used that would be able to sense and change tread characteristics based on the environment [46]. For example, when rougher terrain is encountered, the shoe could become more stiff to offer more stability.



Modular Shoe: The modular shoe solution involves the idea of a two part shoe: the upper and the sole. Uppers have a long lifespan of 1000+ miles or more than a year, whereas soles only last 300-500 miles, or about 3-5 months for the level of runner we are targeting. Accordingly, we devised the two part shoe with the goal of being able to replace soles easily onto the same upper. There could be different soles for different environments; for example, we could make soles for the trail, road, and track. This solution requires the development of a locking mechanism that connects the upper and sole parts of the shoe. Our mechanism will need to be secure, intuitive, and durable.



Recyclable Shoe: Like the ON Running recyclable shoe subscription service, our idea would be 100% recyclable, but because uppers last longer than soles, we would make the shoe so that individual components could be removed and replaced. If the sole or midsole had run its course, it could be replaced without wasting the remaining life of the upper.



Retreadable Material: Given that the sole has a much shorter lifespan than the upper, it is possible that having a way to add treads back into the shoe would be beneficial. This could be done by using a retreadable material that would be able to be attached to worn down shoes and then imprinted with a tread pattern for varying environments. This retreading could come from melting and curing a material and then attaching it to the sole of the shoe. Certain thermoplastics and rubbers are able to be melted at lower temperatures, such as PEA, which melts at 55 degrees celsius [47]. Another method is to utilize putty-like material that is then cured and hardened into the proper shape of the shoe sole [48].

Appendix D: Alternatives Matrix

	Affordability (4)	Versatility (5)	Durability (5)	Performance (4)	Style (2)	Ethical/ legal/ safe (Y/N)	Total
Sole Attachments	3	3	3	3	1	Y	56
Barefoot Pads	4	1	4	2	0	Y	49
Hybrid Shoe	1	3	4	4	2	Y	59
Modular Shoes	3	4	3	3	1	Y	61
100% Recyclable Shoes	2	2	3	2	2	Y	45
Re-treadable Material	4	3	3	2	1	Y	56

Using the specifications we had decided upon, we considered various alternatives during the brainstorming process. In our alternatives matrix, specifications were weighed on a scale of 1 to 5 with 5 as the highest priority. We weighed versatility and durability as the most important factors of the shoe (weighing them a 5), as these factors would be most important in designing an effective and usable shoe. Secondary to these, we valued affordability and performance, as our idea should reduce the need to buy multiple shoes while also being a useful tool for running (weighed these a 4). Style was much less important, as we gave it a weight of 2, but we still wanted to factor it in as we would want runners to be confident in wearing our product. Our idea for a modular shoe scored the highest out of the ideas we compared and accordingly, is the idea we elected to move forward with.

Appendix E: Signifiers Suggested by User Testing

